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Cylinder Head Reuse Guidelines

## Cylinder Head Reuse Guidelines

### Product Affected

- ISX12N CM2380 X120B
- ISX12 G CM2180 EJ

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### I. General Information

The reuse and replacement guidelines are to be followed to reduce unnecessary repairs.

### II. Introduction

- Use the information contained within this document in conjunction with the corresponding Service Manual. Reference Procedure 002-004 in Section 2.
- Do **not** replace cylinder head for valve leakage without identifying and attempting to correct root cause of leakage. In some instances, valve leakage is caused by deposits on valve face

and/or valve seat. These cylinder heads and valves will be acceptable for reuse after lapping.

### III. Valve Numbering

Before cylinder head is disassembled, intake and exhaust valves **must** be numbered as shown in Figure 1 below. See Table 1 below for cylinder head valve numbering. See Figure 2 and Table 2 below for cylinder head combustion face view identification.

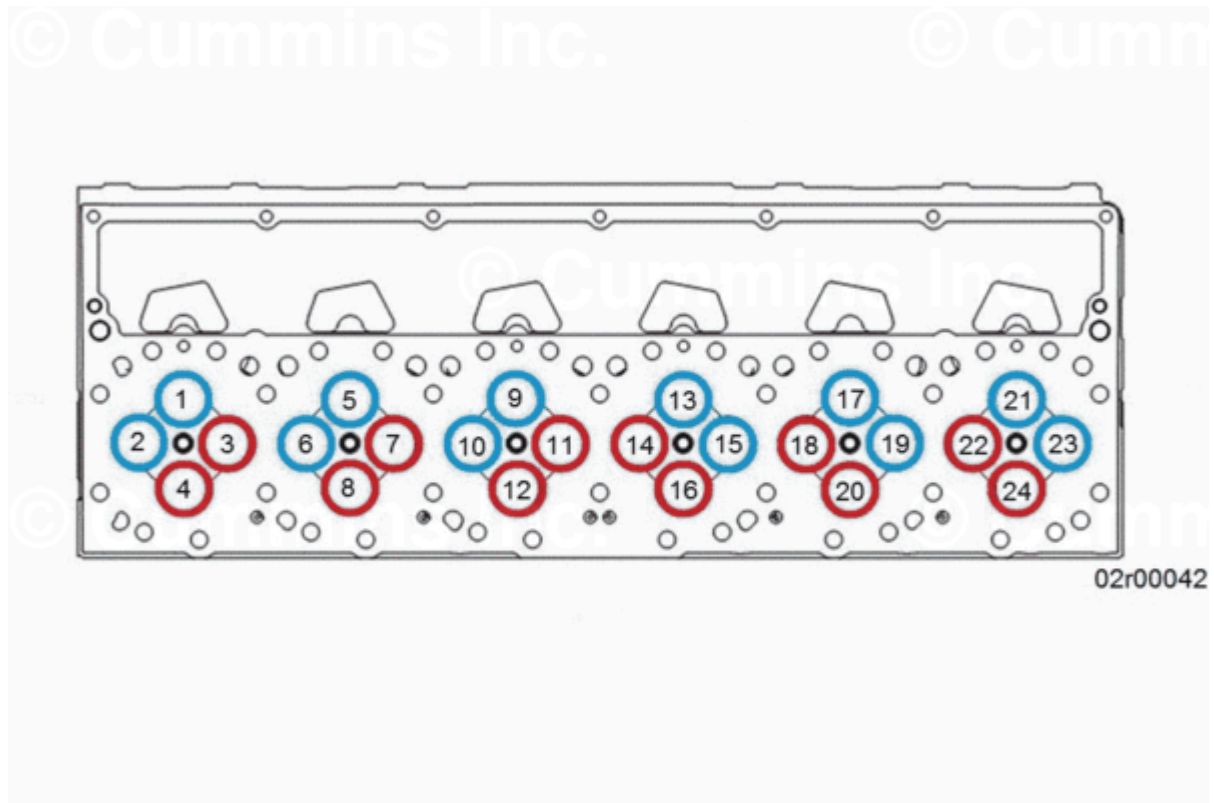


Figure 1 , Cylinder Head Valve Numbering.

**Table 1, Cylinder Head Valve Numbering**

| Figure 1, Item Number                            | Component      |
|--------------------------------------------------|----------------|
| 1, 2, 5, 6, 9, 10, 13, 15, 17, 19, 21, 23 (blue) | Intake Valves  |
| 3, 4, 7, 8, 11, 12, 14, 16, 18, 20, 22, 24 (red) | Exhaust Valves |

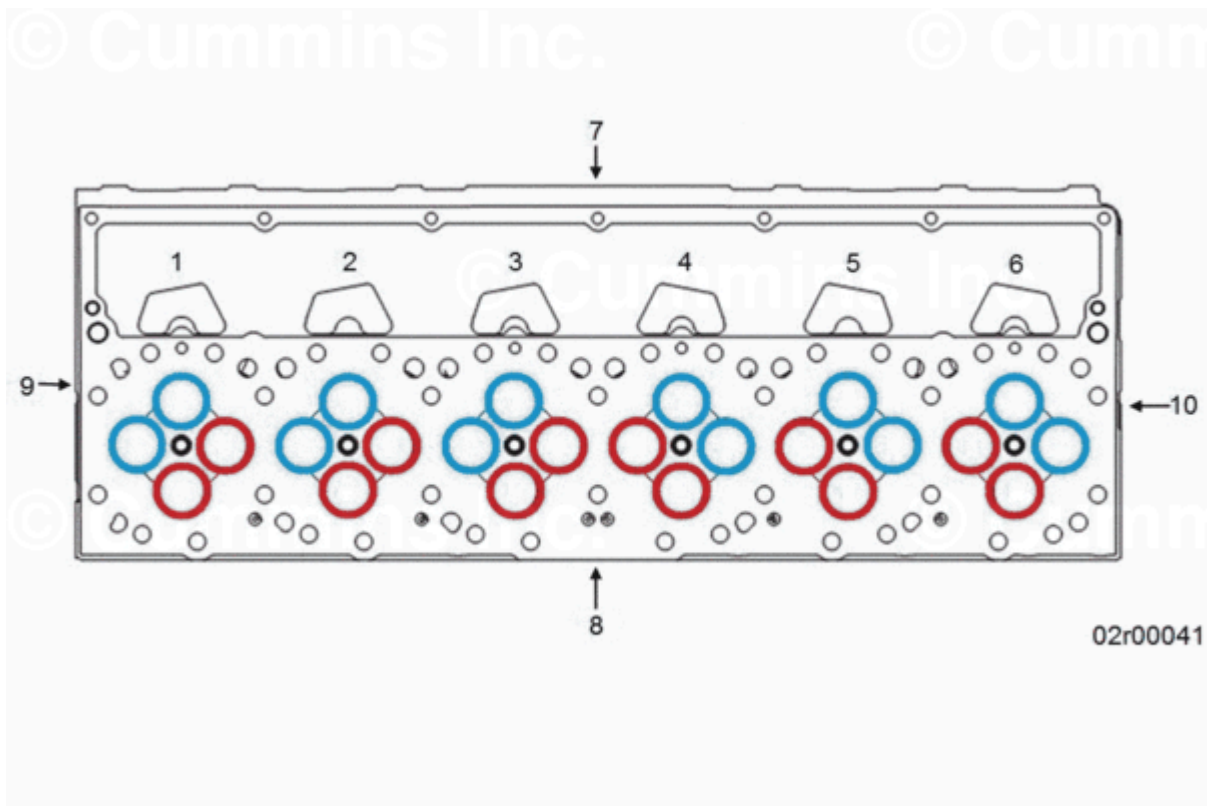


Figure 2, Cylinder Head Combustion Face View.

**Table 2, Cylinder Head Combustion Face View Identification**

| Figure 2, Item Number | Component                     |
|-----------------------|-------------------------------|
| 1, 2, 3, 4, 5, 6      | Cylinder Numbers              |
| 7                     | Intake Side of Cylinder Head  |
| 8                     | Exhaust Side of Cylinder Head |
| 9                     | Front of Engine               |
| 10                    | Rear of Engine                |

#### IV. Valve Terminology

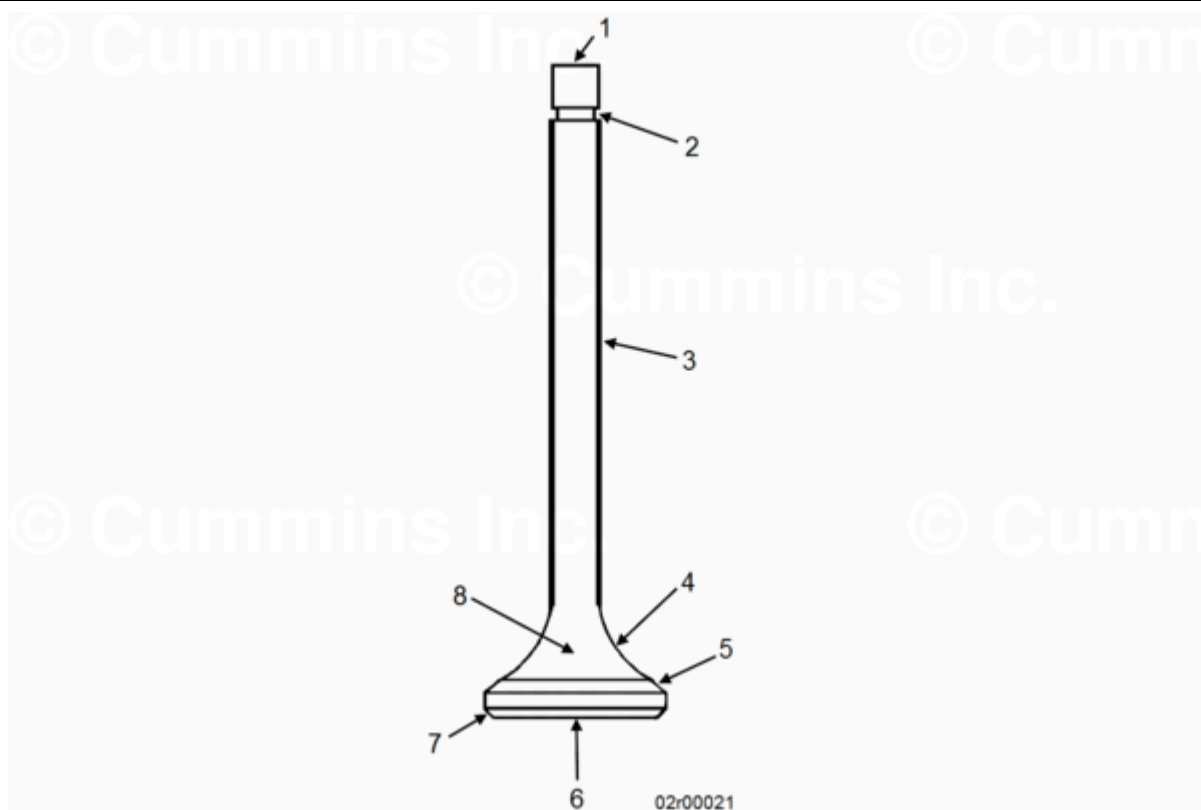


Figure 3, Components of a Valve.

**Table 3, Components of a Valve**

| Figure 3, Item Number | Component             |
|-----------------------|-----------------------|
| 1                     | Tip                   |
| 2                     | Keeper Groove         |
| 3                     | Valve Stem            |
| 4                     | Valve Fillet          |
| 5                     | Valve Seating Face    |
| 6                     | Valve Combustion Face |
| 7                     | Valve Margin          |
| 8                     | Valve Head            |

## V. Inspections

### a. General

Some conditions require removal of valves for cleaning or lapping. Remove valves **only** as directed. All valves **must** be numbered with a paint pen or permanent marker prior to removal. All valves **must** be installed into original location.

### b. Cylinder Head Conditions and Malfunction Modes

#### 1. Combustion Face Deposits

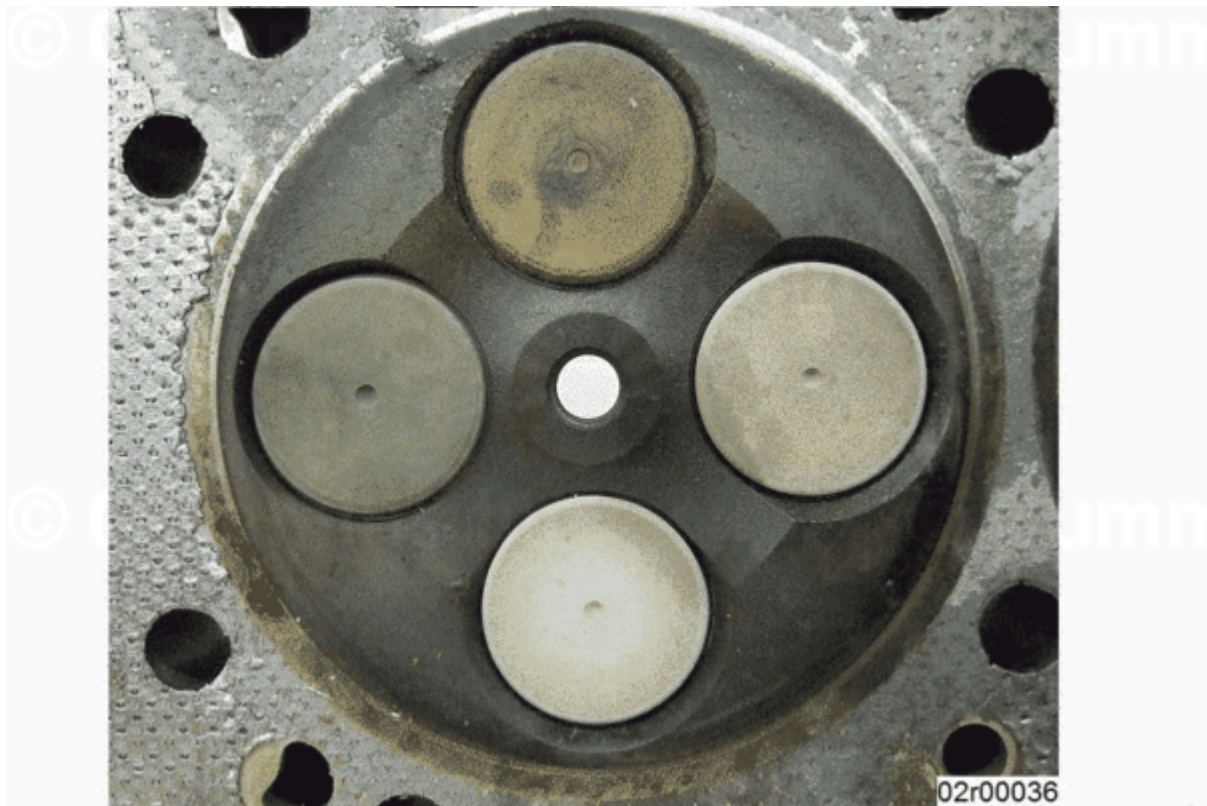


Figure 4, Combustion Face Deposits - Light.

**Table 4, Combustion Face Deposits**

|            |                                                          |
|------------|----------------------------------------------------------|
| Appearance | Light deposits.                                          |
| Cause      | Deposits from combustion and acceptable oil consumption. |
| Action     | Reuse. No cleaning necessary.                            |



Figure 5, Combustion Face Deposits – Moderate.

**Table 5, Combustion Face Deposits**

|            |                                                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Moderate deposits.                                                                                                                                                  |
| Cause      | Deposits from combustion and acceptable oil consumption.                                                                                                            |
| Action     | Conditional Reuse. Clean, inspect, and reuse. Do <b>not</b> remove valves for cleaning. See corresponding Service Manual. Reference Procedure 002-004 in Section 2. |



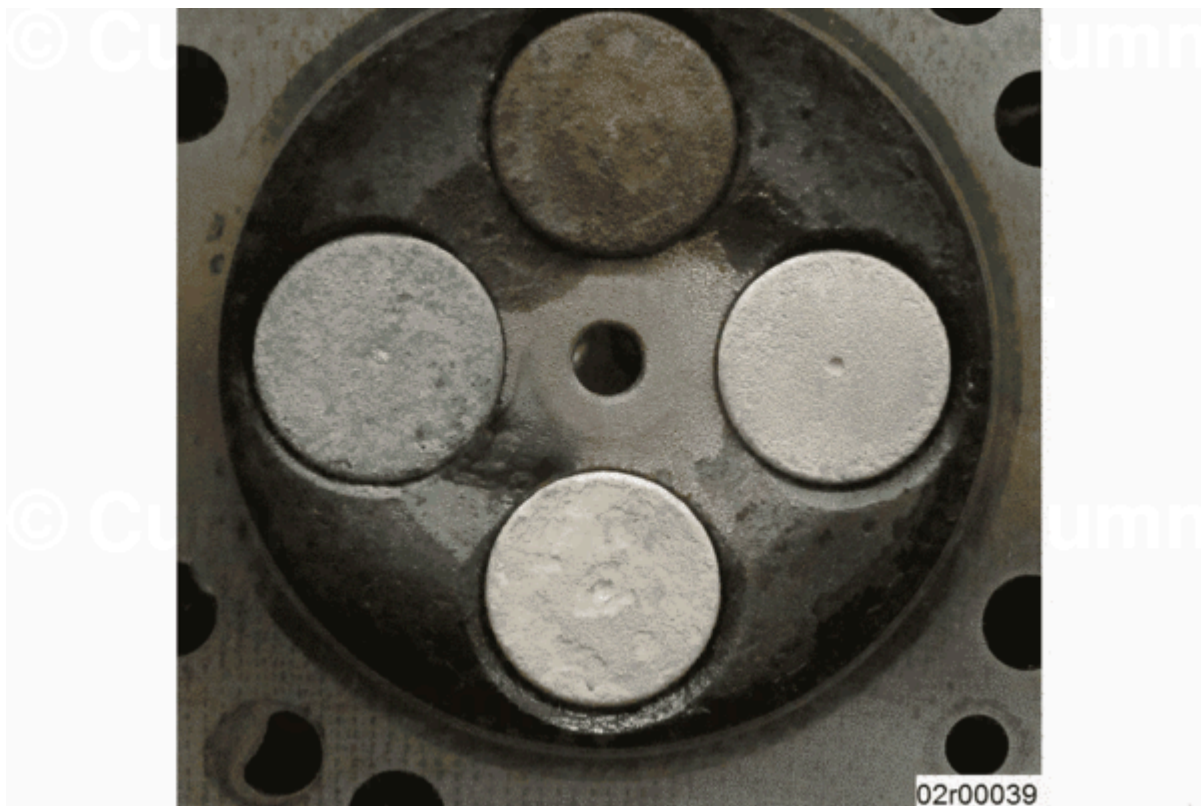


Figure 6, Combustion Face Deposits – Heavy.

**⚠ CAUTION ⚠**

**Caution: Valves must be installed when using a wire wheel or damage can occur.**

**Table 6, Combustion Face Deposits**

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Heavy deposits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Cause      | Deposits from combustion and acceptable oil consumption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Action     | <ul style="list-style-type: none"> <li>• Conditional Reuse.</li> <li>• Clean combustion face and valve head prior to performing any inspections or measurements. This applies to valve leakage, combustion face cracking, valve recession, or flatness.</li> <li>• See corresponding Service Manual. Reference Procedure 002-004 in Section 2.</li> <li>• If necessary, a powered wire wheel or Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, and solvent can be used to remove deposits from cylinder head and valve heads.</li> </ul> |

## 2. Cylinder Head Cracking – Bridge and Spark Plug – Non-Leaking

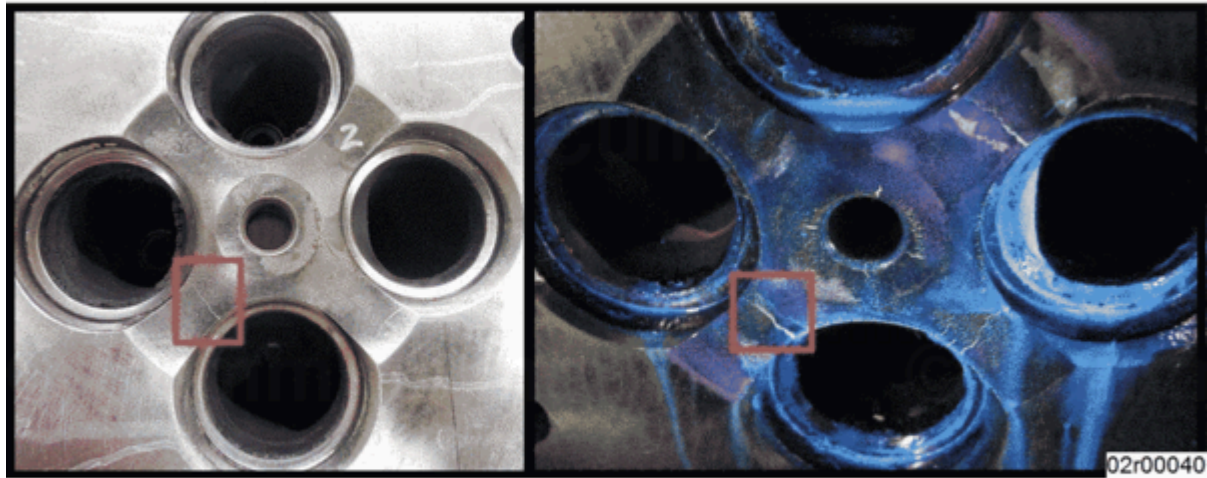


Figure 7, Left: Combustion Face Showing Cracks. Right: Magnafluxed Combustion Face Showing Cracks.

**Table 7, Cylinder Head Combustion Face Cracking – Bridge and Spark Plug – Non-Leaking**

|            |                                                                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Non-leaking crack on combustion face with no crack separation present. All bridges can be cracked and still acceptable for reuse. Reference Cylinder Head Combustion Face Cracking – Leaking for comparison. |
| Cause      | Stress relief crack that does <b>not</b> leak.                                                                                                                                                               |
| Action     | <ul style="list-style-type: none"><li>• Reuse. Clean and inspect cylinder head deck surface.</li><li>• Non-leaking cracking is acceptable at bridge and around spark plugs.</li></ul>                        |

## 3. Cylinder Head Cracking – Leaking



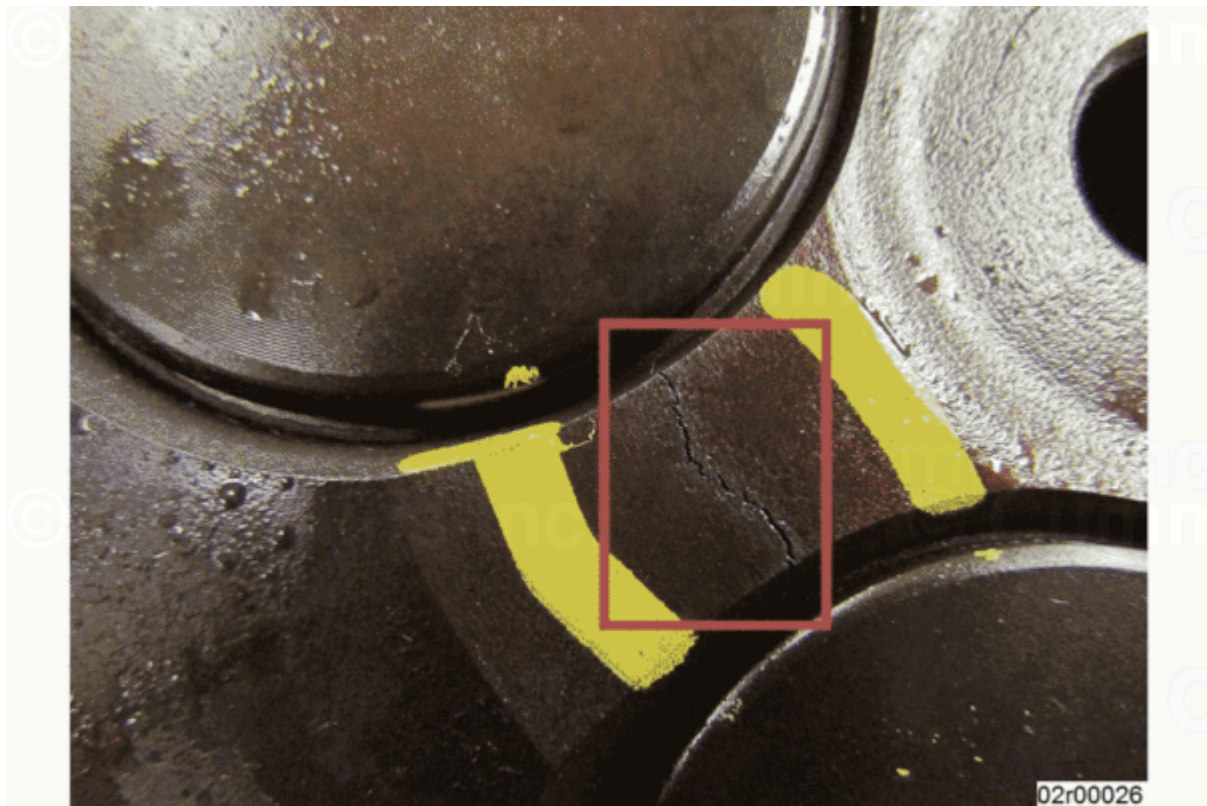


Figure 8, Cylinder Head Combustion Face Cracking - Leaking Bridge.

**Table 8, Cylinder Head Combustion Face Cracking – Leaking Bridge**

|            |                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------|
| Appearance | Leaking crack of combustion face or exhaust port. A leaking bridge crack can be identified by separation.  |
| Cause      | Excessive cylinder temperature and pressure. Likely caused by prolonged operation during knock conditions. |
| Action     | See corresponding Service Manual. Reference Procedure 002-004 in Section 2.                                |

#### 4. Chordal Valve Malfunction

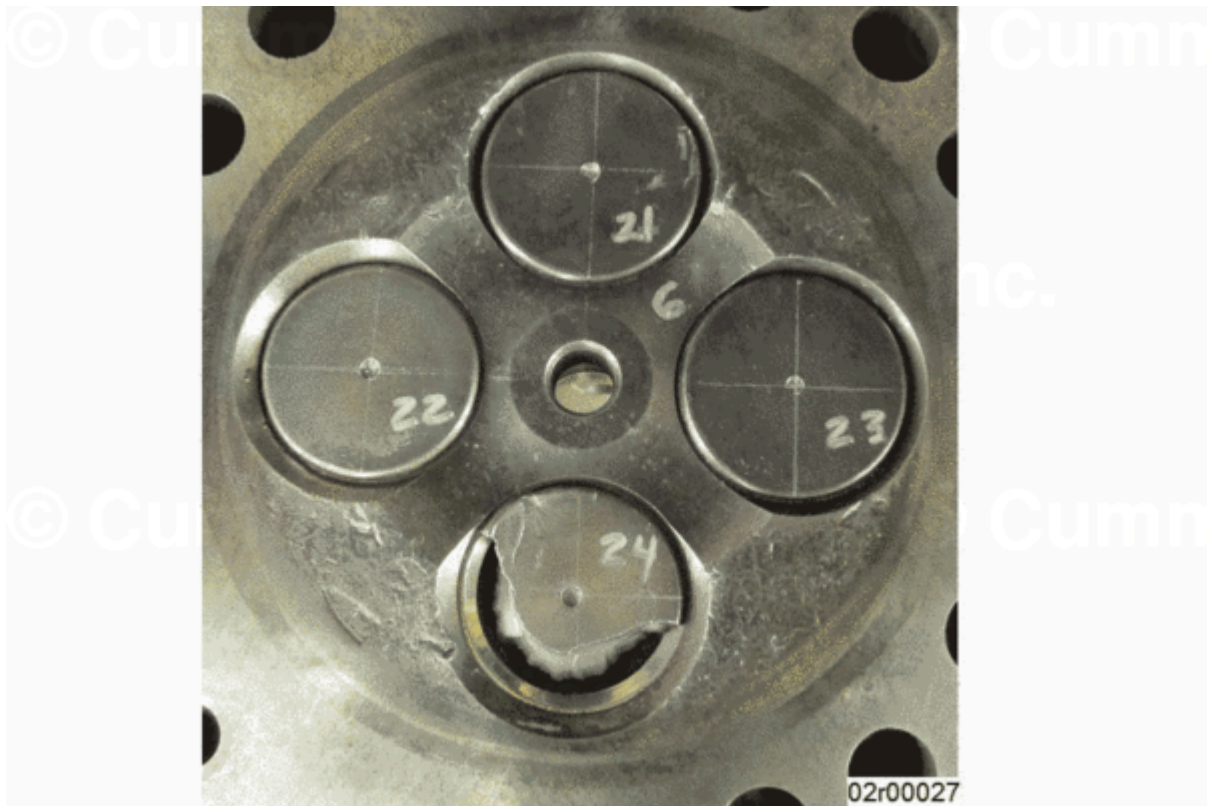


Figure 9, Chordal Valve Malfunction.

**Table 9, Chordal Valve Malfunction**

|            |                                                                             |
|------------|-----------------------------------------------------------------------------|
| Appearance | Clean fracture of valve head.                                               |
| Action     | See corresponding Service Manual. Reference Procedure 002-004 in Section 2. |

## 5. Oil Deposits Trapped in Valve Seat

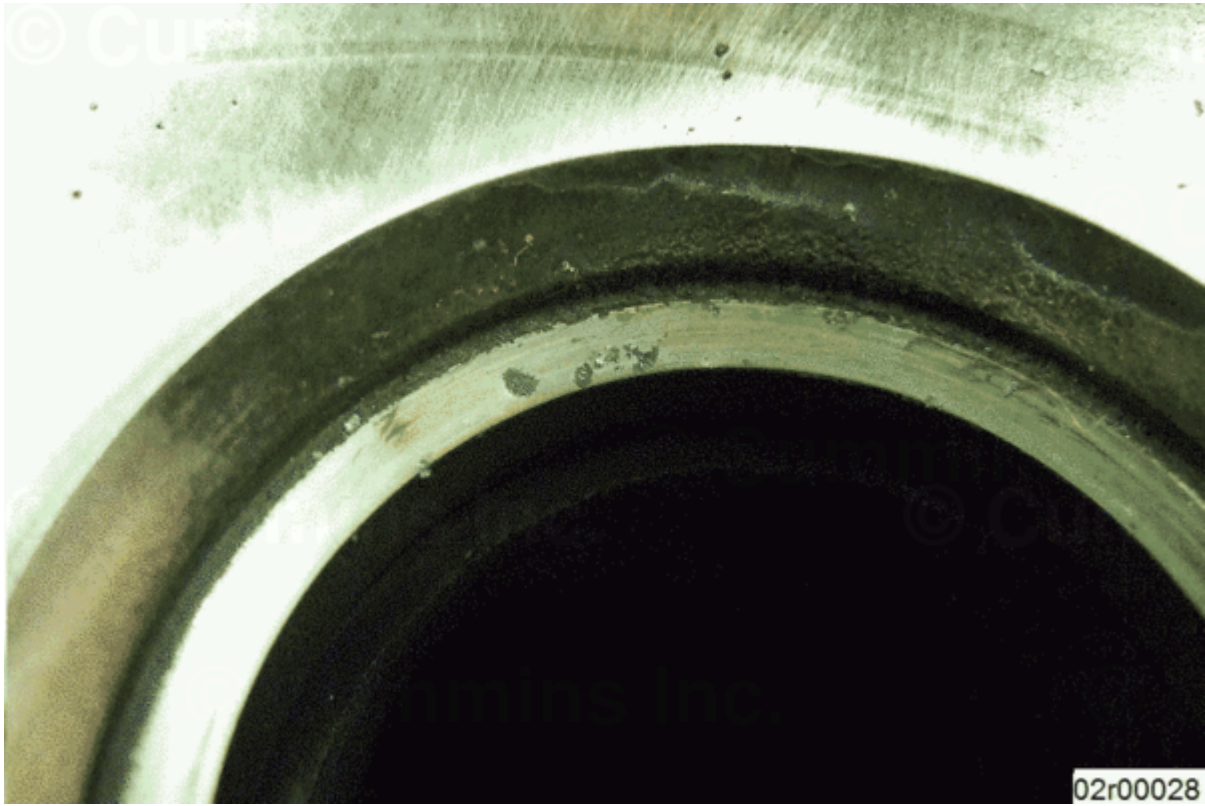


Figure 10, Oil Deposits Trapped in Valve Seat.

| Table 10, Oil Deposits Trapped in Seat |                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance                             | <ul style="list-style-type: none"> <li>• Small deposit visible on valve seat and valve face. Can cause valve to <b>not</b> pass 30 weight oil leak test. Do <b>not</b> replace cylinder head for this observation.</li> <li>• Excessive cylinder leakdown and low compression can be caused by deposits between valve and valve seat.</li> </ul> |
| Cause                                  | Normal deposits can collect on valve seat or valve face. Deposits do <b>not</b> cause any performance issues but can be observed while engine is disassembled for unrelated repair.                                                                                                                                                              |
| Action                                 | Reuse after lapping. See lapping procedure. See corresponding Service Manual. Reference Procedure 002-004 in Section 2.                                                                                                                                                                                                                          |

6. Exhaust Manifold Mounting Surface Damage



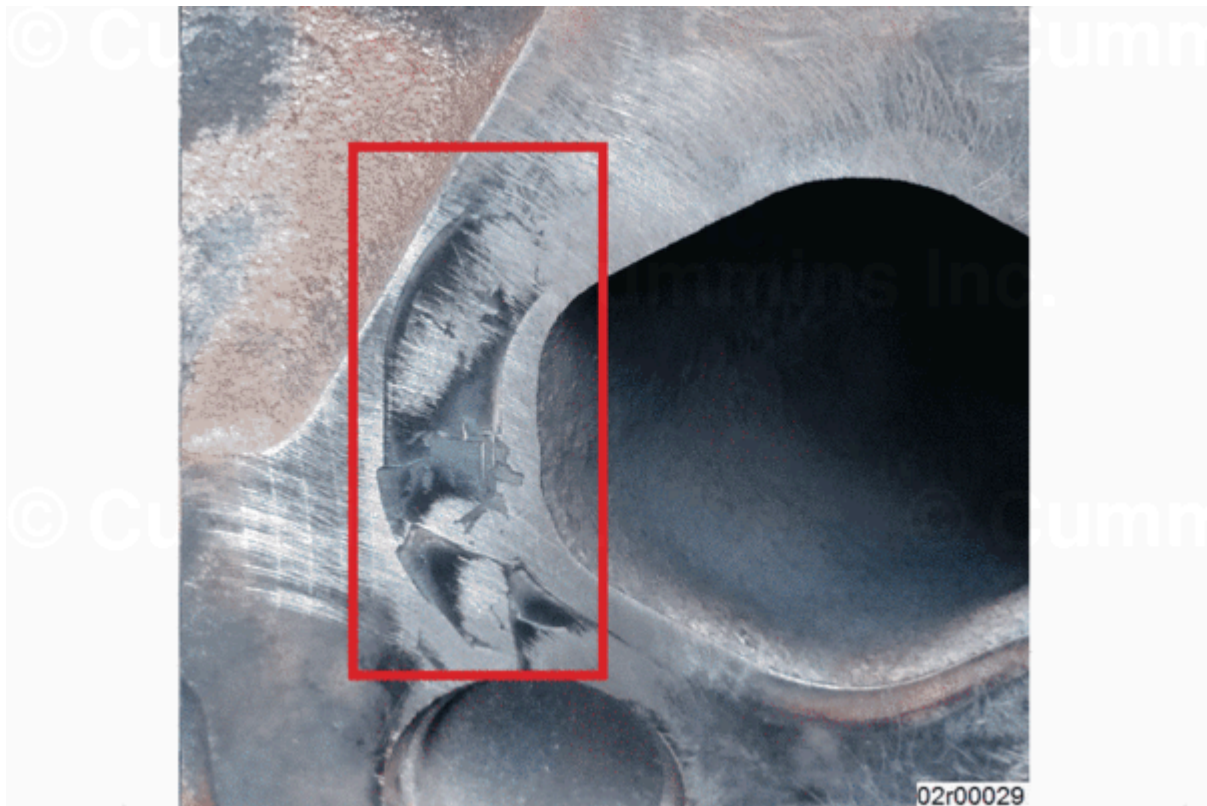


Figure 11, Exhaust Manifold Mounting Surface Damage

**Table 11, Exhaust Manifold Mounting Surface Damage**

|            |                                                                                                                                                                                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Indentation at exhaust manifold mounting surface.                                                                                                                                                                                                                                                                                                       |
| Cause      | Continued operation with leaking exhaust manifold gasket.                                                                                                                                                                                                                                                                                               |
| Action     | <ul style="list-style-type: none"> <li>• Clean gasket sealing area prior to inspection.</li> <li>• Inspect exhaust manifold sealing surface for damage deep enough to be felt with a fingernail. If a fingernail catches, cylinder head <b>must</b> be replaced. See corresponding Service Manual. Reference Procedure 002-004 in Section 2.</li> </ul> |

## 7. Intake Oil Buildup



Figure 12, Left: Oil Buildup Least Severe. Right: Oil Buildup Most Severe.

**Table 12, Oil Buildup on Fillet and Stem of Intake Valve**

|            |                                                                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Coated with black, oily sludge or deposits.                                                                                                                                                                                                                                                 |
| Cause      | <ul style="list-style-type: none"> <li>• All Cylinders: Formation of deposits is dependent on duty cycle. Duty cycles with increased idle are more likely to produce heavier valve fillet deposits.</li> <li>• Single cylinder: Power cylinder issue or leaking valve stem seal.</li> </ul> |
| Action     | <ul style="list-style-type: none"> <li>• Unconditional reuse.</li> <li>• Clean valve with brake clean, abrasive wheel is <b>not</b> to be used.</li> </ul>                                                                                                                                  |

## 8. Asymmetric Valve Seat Contact



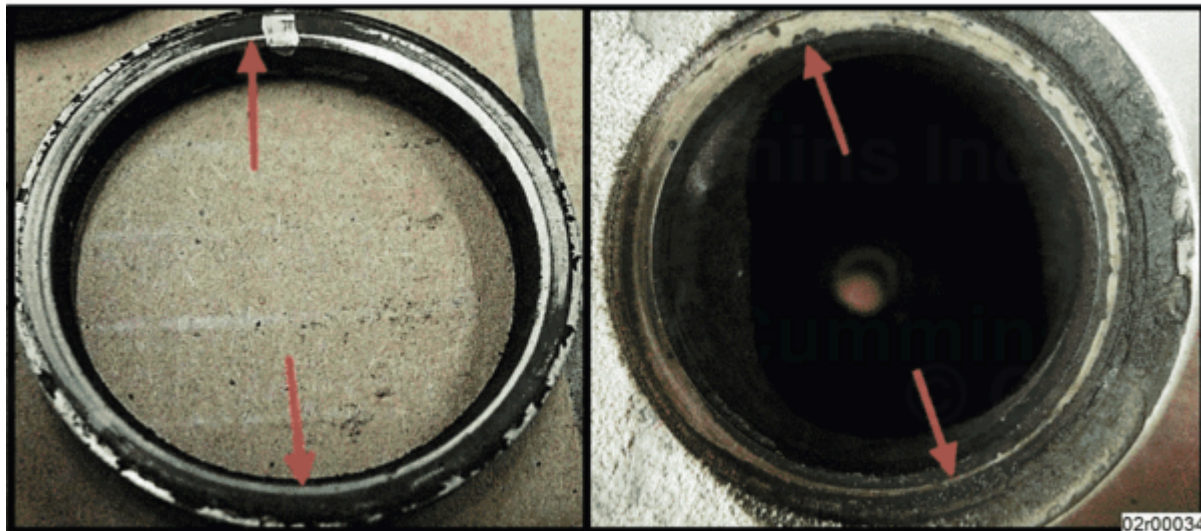


Figure 13, Left: Asymmetric Valve Seat Contact. Right: Asymmetric Valve Seat Contact with Pitting.

**Table 13, Asymmetric Valve Seat Contact**

|            |                                                                                                                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Uneven valve seat contact. Valve can <b>not</b> pass valve leakage test. Asymmetric valve seat contact and deposits can cause low compression and high cylinder leakdown test results.                                 |
| Cause      | <ul style="list-style-type: none"><li>• Reduced valve and seat contact caused by thermal distortion, wear, and deposit buildup.</li><li>• Oil consumption deposits can collect on valve seat and valve face.</li></ul> |
| Action     | <ul style="list-style-type: none"><li>• Reuse after lapping.</li><li>• Perform valve lapping procedure. See corresponding Service Manual. Reference Procedure 002-004 in Section 2.</li></ul>                          |

## 9. Valve Pitting

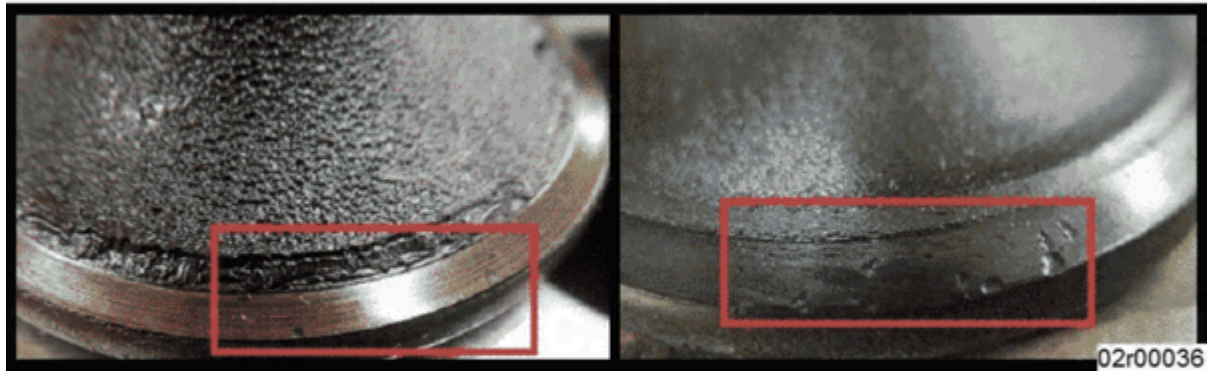


Figure 14, Left: Valve Pitting Least Severe. Right: Valve Pitting Most Severe.

**Table 14, Valve Pitting**

|            |                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Indentation on valve or valve seat faces. Valve can fail valve leakage test. A pitted valve can be reused if valve does <b>not</b> leak.                                                                                                                                                                                                                                      |
| Cause      | Deformation and pitting corrosion caused by power cylinder conditions and deposits collecting on valve and valve seat faces.                                                                                                                                                                                                                                                  |
| Action     | <ul style="list-style-type: none"><li>• Reuse after lapping.</li><li>• Perform valve lapping procedure. See corresponding Service Manual. Reference Procedure 002-004 in Section 2.</li><li>• There is a possibility lapping will <b>not</b> remove all pits from valve and valve seat faces. Valve is to be lapped <b>only</b> until valve passes valve leak test.</li></ul> |

## 10. Torched Exhaust Valve



Figure 15, Torched Exhaust Valve.

**Table 15, Torched Exhaust Valve**

|            |                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | Damaged valve with torch-like cutting appearance.                                                                                             |
| Cause      | Prolonged engine knock.                                                                                                                       |
| Action     | Replace cylinder head and inspect pistons for excessive deposits. See corresponding Service Manual. Reference Procedure 002-004 in Section 2. |

## 11. Burnt Intake Valve



Figure 16, Left: Burnt Intake Valve Least Severe. Right: Burnt Intake Valve Most Severe.

**Table 16, Burnt Intake Valve**

|            |                                                                                                    |
|------------|----------------------------------------------------------------------------------------------------|
| Appearance | Cracking on valve seat face                                                                        |
| Cause      | Thermal fatigue: cylinder head with improved valve and seat materials released 31 March 2015.      |
| Action     | Replace cylinder head. See corresponding Service Manual. Reference Procedure 002-004 in Section 2. |

## 12. Valve Leakage



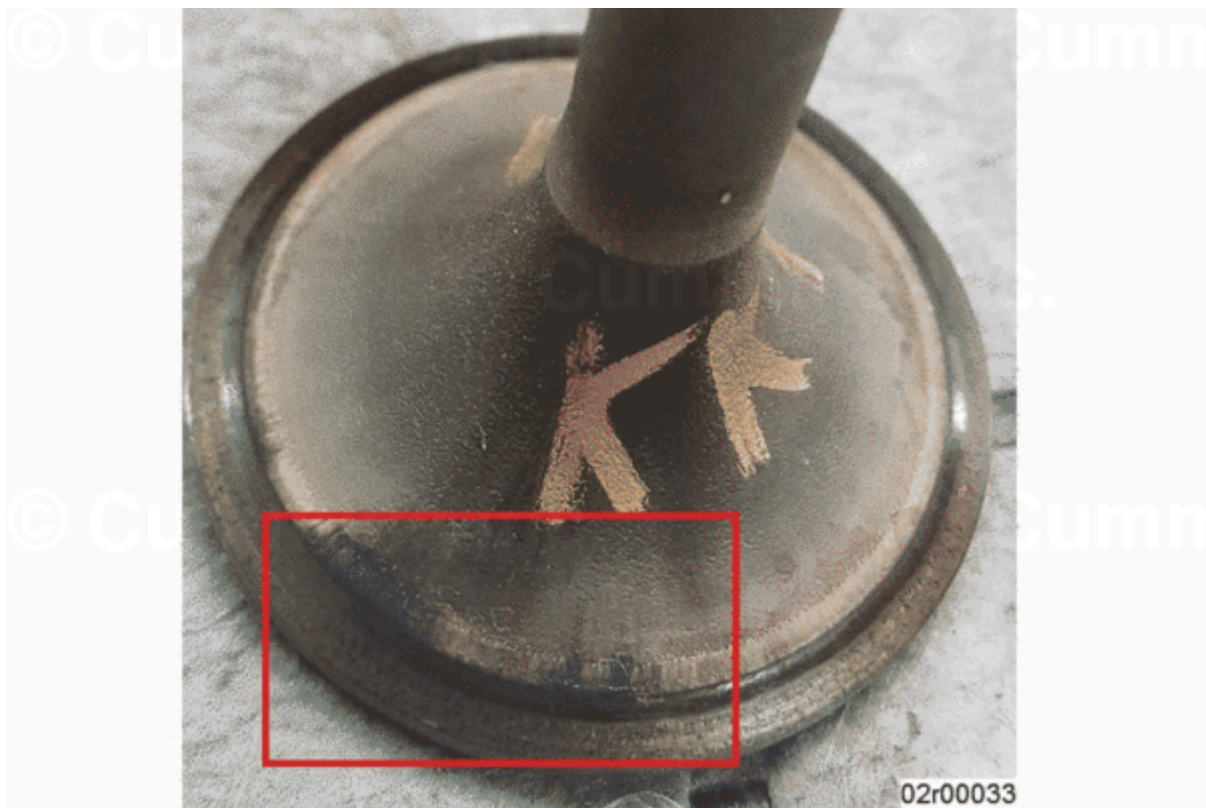


Figure 17, Leaking Valve.

**Table 17, Valve Leakage**

|            |                                                                                                                                                                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance | <ul style="list-style-type: none"> <li>Visual inspection is not able to determine if a valve is leaking. Valve seating contact can appear asymmetric but still seal effectively. Performing valve leak test is only way to identify leakage.</li> <li>A leak forming after 30 seconds is acceptable.</li> </ul> |
| Cause      | <ul style="list-style-type: none"> <li>If no physical valve or seat damage is observed such as burning or torching, valve leakage is likely result of poor contact or a deposit buildup on valve face and valve seat.</li> </ul>                                                                                |
| Action     | Reuse. Reference lapping procedure. See corresponding Service Manual. Reference Procedure 002-004 in Section 2.                                                                                                                                                                                                 |

## VI. Repair

### a. Valve Lapping



Figure 18, Left: Valve Seat and Valve Before Lapping. Right: Valve Seat and Valve After Lapping.

**Table 18, Valve Lapping**

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conditions for Valve Lapping | <p><b>Only</b> lap valves as directed. Lapping will correct all valve leakage except for burning, torching, and chordal malfunction conditions. Lapping will repair sealing surface and clear any debris or imperfections from between valve and valve seat.</p>                                                                                                                                                                                                                     |
| Action                       | <ul style="list-style-type: none"> <li>Lap all four valves of <b>only</b> cylinders that did <b>not</b> pass procedure in corresponding Service Manual. Reference Procedure 002-004 in Section 2.</li> <li>Number all valves and remove all four valves from only cylinder that did <b>not</b> pass valve leak test. Use valve spring compressor tool, Part Number 5298756, to remove valves. See corresponding Service Manual. Reference Procedure 002-004 in Section 2.</li> </ul> |

# Document History

| Date      | Details                                              |
|-----------|------------------------------------------------------|
| 2017-2-15 | Module Created                                       |
| 2018-4-30 | Updated text to reference Figure 3 in Table 3.       |
| 2021-6-14 | Correcting error that causes graphic to not show up. |
| 2021-9-13 | Added Product Affected Section.                      |
| 2023-2-27 | Fixed heading format throughout service bulletin.    |

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**Last Modified: 27-Feb-2023**

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